

Louay Guetat

AI ENGINEER

Tunis, Tunisia · Open to relocation worldwide

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AI Engineer specializing in multi-agent systems, LLM integration, and RAG pipelines, backed by a strong full-stack foundation. I build production AI platforms end to end — from retrieval and agent orchestration to the APIs and UIs that ship them — with an emphasis on evaluation, guardrails, and reliable, source-grounded output. Currently building financial research agents and an AI diligence platform for private markets.

WORK EXPERIENCE

AI Engineer — Alpha Momentum

Remote · US Startup | February 2025 – Present

Led AI/ML development for a financial research platform built on multi-agent systems, RAG pipelines, and intelligent data processing, while owning the surrounding full-stack infrastructure (React web, React Native mobile, serverless Node.js API). Since August 2026 I have also led Capital Thread, an AI diligence platform for private markets (detailed under Selected Projects), through Alpha Momentum.

- Architected a multi-agent system with LangChain and LangGraph — specialized agents for equity research, valuation, portfolio optimization, risk, technical analysis, and market sentiment — integrating multiple LLM providers (OpenAI, Anthropic, Google Gemini, xAI, Groq, SambaNova, Ollama) with dynamic selection and fallback
- Developed Dexter, a financial research agent with custom tools for financial search, SEC filing analysis, web search (Tavily/Exa), and browser automation (Playwright), plus a markdown-based skills system for specialized analyses (DCF models, options) with frontmatter metadata
- Built an evaluation framework with rubric-based assessment, numerical-accuracy validation, tool-selection analysis, and confidence intervals, using LangSmith for tracing and performance monitoring across agent iterations
- Implemented a RAG pipeline with intelligent chunking, OpenAI embeddings, Neo4j graph storage for semantic relationships, and optimized retrieval over financial documents for context-aware responses
- Built FastAPI services for SEC filing processing (edgartools + custom parsers) extracting structured data from 10-K, 10-Q, and 8-K filings, with AWS Batch orchestration for 37k+ documents and incremental updates
- Built Python data-ingestion pipelines (BeautifulSoup, Selenium, aiohttp) for FMP, Federal Reserve, ECB, news, earnings transcripts, and market signals, with structured logging and error handling
- Engineered the full-stack delivery layer: a React 18 + TypeScript web app with real-time AI streaming (SSE), a serverless Node.js API with LangChain integration, React Native apps, and GitHub Actions CI/CD to Docker/AWS

Tech Stack: Python (FastAPI, LangChain, LangGraph, LangSmith, Pandas, NumPy), OpenAI, Anthropic, Google Gemini, xAI, Groq, Neo4j, MongoDB, Redis, Docker, AWS (Lambda, ECS, ECR, Batch), edgartools, BeautifulSoup, Selenium, Playwright, TypeScript, React, Node.js

Data Science Intern — Sagemcom

Tunis | February 2024 – August 2024

- Built multi-agent systems with LLM integration and embedding-based retrieval
- Built an end-to-end data pipeline — scraping and dataset assembly (Selenium, BeautifulSoup), retrieval with HuggingFace models and LangChain, and automated workflows with LangGraph
- Developed a ReactJS + Flask web interface for model interactions over a REST API

Technologies: Python, Flask, LangChain, LangGraph, HuggingFace, ReactJS, SQLite, REST API

Full Stack Developer — WeTekup

Tunis | March 2025 – September 2025

- Built responsive React websites (SCSS) with Node.js/Express REST APIs over PostgreSQL
- Deployed to production on Hostinger with an Nginx reverse proxy, SSL/TLS, and performance tuning

Technologies: React, Node.js, Express.js, SCSS, PostgreSQL, Nginx

Full Stack Intern — WeTekup (Khedma.site)

Tunis | July 2023 – September 2023

- Built a job-marketplace platform with a ReactJS front end and Django REST APIs over PostgreSQL

Technologies: ReactJS, Django, Python, PostgreSQL

SELECTED PROJECTS

Capital Thread — AI Diligence Platform for Private Markets · *Client engagement via Alpha Momentum* · Aug 2026 – Present

One-click diligence pipeline for private credit, PE, and venture: sources deals from SEC EDGAR and Form D, parses Inline XBRL, and generates financial models, DCF valuations, and IC memos where every figure deep-links to its source filing. Architected so LLMs (OpenAI, Anthropic) handle judgment and narrative while deterministic Python handles all arithmetic — grounding output and eliminating hallucinated numbers. 437 unit tests on the financial engines.

Tech: OpenAI API, Anthropic API, FastAPI, PostgreSQL, SQLAlchemy, SEC EDGAR, Arelle XBRL, React 19, TypeScript

Alpha Momentum — Multi-Agent Financial Research · *alphamomentum.ai* · Live

Five LangGraph research workflows and a RAG pipeline (OpenAI embeddings + Neo4j) over a 37k+ document SEC corpus, delivered across web and mobile.

Tech: LangChain, LangGraph, OpenAI, Anthropic, Neo4j, FastAPI, AWS Batch

Intellectia — Vision-Powered Math Learning · *Team project*

Math learning platform for children that turns text problems into visual representations using vision models, with interactive quizzes and performance dashboards.

Tech: TensorFlow, HuggingFace, React, Django, MongoDB, NumPy, Pandas

PMBOK Risk Recommender — NLP / GNN · *Team project*

Chatbot-style project-risk recommender trained on PMBOK 11th-edition data, using a Graph Attention Network with a BERT embedder for text-based recommendations.

Tech: PyTorch, TensorFlow, BERT, GNN, NLP, Django, NumPy

Yoga Pose Classifier — Computer Vision · *Solo build*

CNN-based yoga pose classification; a custom CNN reached 86% accuracy and ResNet reached 97% across multiple pose categories.

Tech: Python, CNN, ResNet, Django, NumPy, Pandas

TECHNICAL SKILLS

AI/ML & LLM: LangChain, LangGraph, LangSmith, OpenAI API, Anthropic API, Google Gemini, RAG, HuggingFace, multi-agent systems, prompt/agent evaluation

Deep Learning: TensorFlow, PyTorch, Keras, scikit-learn, CNN, ResNet, BERT, GNN, NLP, Computer Vision

Data: Pandas, NumPy, BeautifulSoup, Selenium, Playwright, aiohttp, SEC EDGAR, Arelle XBRL

Backend: Python (FastAPI, Flask, Django), Node.js (Express, Serverless)

Databases: PostgreSQL, MongoDB, Neo4j, MySQL, Redis, DynamoDB

Cloud & DevOps: AWS (Lambda, ECS, ECR, Batch, S3, CloudFront, Cognito), Docker, GitHub Actions

Frontend: React 19, TypeScript, React Native, Next.js, MUI, Tailwind CSS

EDUCATION

Computer Engineering, Data Science — ESPRIT

Tunis · 2021 – 2024

- Thesis: “Generative AI with Business Knowledge” — Mention: Very Good

Bachelor of Computer Science — ESSECT

Tunis · 2018 – 2021

- Foundation in software development, algorithms, and data structures

ADDITIONAL INFORMATION

- **Languages:** Arabic (Native), English (B2, Professional), French (B2, Professional)
- **Soft skills:** Problem-solving, collaboration, and clear communication